# 01 SIGNAL CIRCUITS

A.9 / WIRED READING VIEW  Continuous signal wiring | dots mark connections | six supply rails link the sheets

## 1 TCXO / CAPTURE CLOCK

TP9

3V3_TCXO

X1
ECS-TXO-5032-160-TR
VDD
NC
OUT
GND
ECS_RAW

TP1

U1B
74AHCT2G125DP
ECS_BUF

R2
33R

TP2
CLK_NANO

5V_CAPTURE

U1C
74AHCT2G125DP
VCC
GND

C1
100n

C2
1u

TP8

NANO_VIN

A1
Arduino_Nano_Every
VIN
+5V
D2/CLOCK

C3
100n

C4
4.7u

## 2 PENDULUM / OPTICAL INPUT

JP7
LED RESISTOR BYPASS

JP5
OPT PULLUP

PPS default: JP9 + JP10 both 2–3.
Direct: both 1–2. JP11 open.

R7
10k

R3
150R

J1
PENDULUM / SCREW
RED_LED+
BLACK_LED−
WHITE_VCC
BLUE_OUT
GREEN_GND
OPT_J1

OPTO_5V

J3
OPTO TRS
S
R
SN
T
STN
OPT_JACK

R6
100R

OPT_SELECTED

TP3

U3
74AHCT1G176V
VCC
A
NC
GND
Y

R4
0R

TP4

OPT_NANO

D12/OPTICAL

D1
TPD2E2U06DCKR

C5
100n

C6
1u

C11
100n

TP5

GNSS_VCC

J5
GNSS / SELECTED VCC

J4
EXT PPS / DETECT
SLEEVE
TIP
RING
PPS_NC
PPS_COM
PPS_NO
DET_NC
DET_COM
DET_NO

R8
100R

JP10
Buffer input

JP9
PPS source

TP7

PPS_NANO

A3/PPS

D3/REF_DETECT

## 3 GNSS / EXTERNAL PPS

PPS_RAW

5V_IN
PPS_OUT_3V3
TX_OUT_3V3
RX_IN_3V3
GND

PPS_SELECTED

PPS_A

U1A
74AHCT2G125DP

TP6

PPS_BUF

R5
10k

JP11
PPS BIAS

PPS_5V

JP6
REF PULLUP

## 4 UART LEVEL TRANSLATION

U5
TXU0202DCUR
VCCA_5V
VCCB_3V3
A1_/_NANO_TX
A2Y_/_NANO_RX
OE
B1Y_/_NANO_TX_3V3
B2_/_GNSS_TX
GND

NANO_RX5

NANO_TX5

R11
47k

R12
47k

R15
1k

R14
10k

3V3_NANO

3V3/UART_VCC

TX1/UART_TX

RX1/UART_RX

GND

JP8
NANO UART ENABLE

UART_ENABLE

R13
100k

TP10

J2
GNSS MONITOR
GND
GNSS_PPS
GNSS_TX

EXT_PULLUP_FEED

R16
10k

R17
100R

GPS_TX

U6
74AHCT1G17GVH
VCC
A
NC
GND
Y
EXT_PPS_CLEAN

C17
100n

C7
100n

C8
1u

C14
100n

C15
100n

FUNCTIONAL
PIN VIEW
RESET
D4
D5
D6
D7
D8
D10
MOSI
MISO
SCK
AREF
A0
A1
A2
SDA/A4
SCL/A5
A6
A7
RESET

02 POWER CIRCUITS

USB-C and optional 5V input protection
Boost / Nano VIN / accessory limits
Dedicated TCXO and GNSS regulators

File: Wired_Power.kicad_sch

SIGNAL: continuous wires; dots join wires. Crossings without dots are separate.
Six named supply rails continue on sheet 02. Ground symbols share the common return.

Reading view: manufacturing source unchanged; unbuilt prototype
**Pendulum observatory**
Sheet: /
File: NanoEveryCarrier-A9-Wired.kicad_sch

**Title: A.9 wired — SIGNAL CIRCUITS**

| Size: A2 | Date: 2026-11-06 | **Rev: A.9 wired** |
|---|---|---|
| KiCad E.D.A. 10.0.4 | | Id: 1/2 |

# 02 POWER CIRCUITS

A.9 / WIRED READING VIEW  Continuous signal wiring | dots mark connections | six supply rails link the sheets

## 1 USB-C SOURCE QUALIFICATION

J6: 5V sink only.
1.5A / 3A CC required.
No PD: hardware qualification.

U8
STUSB4500LQTR
C20
100n
J6
USB-C / 5V POWER
CC1
R18
2.2k
USB_C_VBUS
USB_VBUS_SENSE
CC_VREG_1V2
CC_VREG_2V7
C18
1u
C19
1u

## 2 INDEPENDENT INPUT PROTECTION

U11
TPS259470ARPWR
USB_EFUSE_EN
BOOST_5V
R23
357k
USB_OVLO
R24
100k
R25
3.32k
USB_DVDT
C24
1n

J7 optional / DNP.
4.75–5.25V regulated.
Inputs reverse blocked:
no fixed source priority.

J7
OPTIONAL 5V INPUT
U12
TPS259470ARPWR
TERM_5V_IN
R26
274k
TERM_UVLO
R27
100k
C26
1u
R28
357k
TERM_OVLO
R29
100k
TERM_ILIM
R30
3.32k
C25
1n

U9
TPS70933DBVR
C22
1u
C21
4.7u
CC_LOGIC_3V3
R20
47k
R19
47k
USB_ATTACH_OK_N
U10
SN74LVC1G02DBVR
R21
10k
USB_POWER_ALLOWED
C23
100n
R22
15k

## 3 ACCESSORY SUPPLIES

5V_CAPTURE
C9
1u
U2
TPS2553DBVR
JP3
OPTO POWER
OPTO_5V
C10
1u
JACK_ILIM
R9
232k
JACK_5V_LIMITED
JP4
PPS POWER
PPS_5V
C12
1u
U4
TPS2553DBVR
C33
1u
U14
TLV1117LV33DCYR
GNSS_5V_LIMITED
GNSS_3V3
JP2
GNSS VCC / 3V3–5V
C13
1u
C34
4.7u
GNSS_ILIM
R10
88.7k
GNSS_VCC

## 4 BOOST / BLOCKING / NANO VIN

L1
4.7uH / 2.5A
D2
SS24
D3
SS14
JP1
EXT VIN ENABLE
EXT_VIN_FEED
NANO_VIN
C27
10u
C31
1u
U13
TPS61085PWR
BOOST_SW
C30
22u
BOOST_8V
R31
100k
C32
22u
BOOST_COMP
R33
13k
C28
3.3n
BOOST_FB
R32
18.2k
BOOST_SS
C29
100n

## 5 QUIET TCXO SUPPLY

R1
22R
C16
10u
U7
TPS7A2033PDBVR
TCXO_LDO_IN
3V3_TCXO

POWER: six named supply rails continue on sheet 01. TCXO supply is dedicated to X1.
Unbuilt prototype: analogue performance, thermal behaviour and power-transfer continuity require bench acceptance.

Reading view: manufacturing source unchanged; unbuilt prototype
**Pendulum observatory**
Sheet: /02 POWER CIRCUITS/
File: Wired_Power.kicad_sch

***Title: A.9 wired — POWER CIRCUITS***

| Size: A2 | Date: 2026-11-06 | **Rev: A.9 wired** |
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